

A Structural Theory of Convenience Yield

Dimensional Constraints, Nation–Currency Architecture, and the Shadow Price of Physical Security

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Abstract

Convenience yield is traditionally interpreted as the non-monetary benefit obtained from holding a physical asset rather than a financial claim on that asset. Existing literature typically decomposes convenience yield into numerous operational, strategic, informational, and financial components. This paper proposes a structural reinterpretation based upon a bounded nation–currency architecture.

Assuming a maximum number of nations $N_{\max} \leq 4$ and a maximum number of reserve currencies $R_{\max} = 5$, we show that the dimensionality of convenience-yield generation is bounded above by ten nation–currency relationships. Consequently, large phenomenological decompositions cannot generally constitute fundamental bases. We distinguish between phenomenological decompositions and structural decompositions and derive a six-factor representation consistent with dimensional constraints.

The framework connects economics, finance, financial economics, welfare economics, political science, international relations, warfare economics, psychology, philosophy, data science, artificial intelligence, and systems theory through a common notion of scarcity-mediated shadow prices.

The paper ends with “The End”

Contents

1 Introduction	1
2 Phenomenological Convenience Yield	2
3 Structural Nation–Currency Architecture	2
4 Dimensionality Theorem	3
5 Incompatibility of Fourteen-Dimensional Bases	3
6 The Six-Factor Structural Basis	3
7 Dimensionality Reduction	4
8 Factor Mapping	4
9 Psychological Interpretation	4
10 Philosophical Interpretation	5
11 Political Science and International Relations	5
12 Warfare Economics	5
13 Artificial Intelligence Interpretation	5
14 Algorithm	6

15 Statistical Framework	6
16 Network Representation	7
17 Conclusion	7

List of Figures

1 Bounded nation–currency architecture	7
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List of Tables

1 Mapping from phenomenological factors to structural basis factors.	4
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1 Introduction

Convenience yield occupies a central role in commodity pricing, inventory management, financial economics, and strategic resource allocation.

The classical cost-of-carry relationship is

$$F_t = S_t e^{(r+u-y)(T-t)},$$

where

- F_t denotes the futures price,
- S_t denotes the spot price,
- r denotes the financing rate,
- u denotes storage cost,
- y denotes convenience yield.

Traditionally, convenience yield is viewed as the collection of all benefits associated with physical possession.

This paper argues that such decompositions may overstate the true dimensionality of convenience-yield generation.

2 Phenomenological Convenience Yield

A common decomposition is

$$y = \sum_{i=1}^{14} y_i.$$

The fourteen components may include:

1. Production continuity,
2. Stockout avoidance,
3. Supply-chain reliability,
4. Strategic security,
5. Operational flexibility,
6. Market power,
7. Timing option value,

8. Informational advantage,
9. Liquidity assurance,
10. Contract fulfillment,
11. Strategic reserves,
12. Speculative opportunity,
13. Geopolitical influence,
14. Financial stability.

Such a decomposition is descriptive rather than structural.
At any given time,

$$y_i(t) = 0$$

may hold for many components.

Consequently, the existence of fourteen categories does not imply fourteen independent dimensions.

3 Structural Nation–Currency Architecture

We introduce the structural constraints

$$N_{\max} \leq 4,$$

and

$$R_{\max} = 5.$$

Definition 1. *A nation–currency pair is a distinct interaction between a sovereign nation and a reserve-currency system capable of generating scarcity-dependent convenience yield.*

The maximum number of independent generators is therefore bounded by

$$D_{\max} = \frac{N_{\max} R_{\max}}{2}.$$

Substituting the bounds gives

$$D_{\max} = \frac{4 \times 5}{2} = 10.$$

4 Dimensionality Theorem

Theorem 1. *If convenience yield is generated solely through nation–currency interactions and if*

$$N_{\max} \leq 4, \quad R_{\max} = 5,$$

then the dimensionality of convenience-yield generation satisfies

$$\dim(\mathcal{Y}) \leq 10.$$

Proof. Each independent generator must correspond to a distinct nation–currency interaction.

The total number of admissible generators is bounded above by

$$\frac{N_{\max} R_{\max}}{2}.$$

Substituting the structural constraints yields

$$\dim(\mathcal{Y}) \leq \frac{4 \times 5}{2} = 10.$$

Therefore

$$\dim(\mathcal{Y}) \leq 10.$$

□

5 Incompatibility of Fourteen-Dimensional Bases

Proposition 1. *A fourteen-factor decomposition cannot generally constitute a fundamental basis under the nation–currency architecture.*

Proof. Assume a fourteen-dimensional basis exists.

Then

$$\dim(\mathcal{Y}) \geq 14.$$

However,

$$\dim(\mathcal{Y}) \leq 10.$$

Thus

$$14 \leq 10,$$

which is impossible.

Therefore at least four factors must be redundant, dependent, emergent, or regime-specific.

□

6 The Six-Factor Structural Basis

We propose

$$y = y_O + y_P + y_R + y_S + y_I + y_G,$$

where

$$\begin{aligned} y_O &= \text{Operational Yield,} \\ y_P &= \text{Option Yield,} \\ y_R &= \text{Risk Yield,} \\ y_S &= \text{Strategic Yield,} \\ y_I &= \text{Informational Yield,} \\ y_G &= \text{Political Yield.} \end{aligned}$$

Since

$$6 < 10,$$

the decomposition is dimensionally feasible.

7 Dimensionality Reduction

The phenomenological representation

$$\mathbf{y}_{14} \in \mathbb{R}^{14}$$

is projected into

$$\mathbf{y}_6 \in \mathbb{R}^6.$$

Define

$$\mathbf{y}_6 = A\mathbf{y}_{14},$$

where

$$A \in \mathbb{R}^{6 \times 14}.$$

This is analogous to

- Principal Component Analysis,
- Autoencoder compression,
- Latent-factor extraction,
- Representation learning.

8 Factor Mapping

Detailed Factor	Structural Category
Production Continuity	Operational
Stockout Avoidance	Risk
Supply Reliability	Risk
Strategic Security	Strategic
Flexibility	Option
Market Power	Strategic
Timing Option	Option
Information	Informational
Liquidity Assurance	Risk
Contract Fulfillment	Operational
Strategic Reserve	Strategic
Speculation	Option
Geopolitics	Political
Financial Stability	Risk

Table 1: Mapping from phenomenological factors to structural basis factors.

9 Psychological Interpretation

Inventory ownership affects perceived uncertainty.

Let

$$U = f(\sigma, Q),$$

where

- σ denotes uncertainty,
- Q denotes inventory.

Then

$$\frac{\partial U}{\partial Q} < 0.$$

Convenience yield therefore reflects a reduction in perceived scarcity anxiety.

10 Philosophical Interpretation

Convenience yield may be interpreted as the value of optionality under uncertainty.

Ownership confers agency.

Agency generates flexibility.

Flexibility generates utility.

Therefore convenience yield becomes a measurable manifestation of freedom under scarcity.

11 Political Science and International Relations

States hold inventories because strategic reserves provide geopolitical leverage.

Let

$$G = g(R_s),$$

where R_s denotes strategic reserves.

Then

$$\frac{\partial G}{\partial R_s} > 0.$$

Convenience yield therefore acquires a geopolitical component beyond market valuation.

12 Warfare Economics

During conflict,

$$y_S \gg y_O.$$

The strategic component dominates operational considerations.

Fuel, food, metals, and ammunition inventories become national-security assets.

Convenience yield becomes a shadow price of military readiness.

13 Artificial Intelligence Interpretation

Let

$$X \in \mathbb{R}^{14}$$

represent the detailed-factor space.

An encoder

$$E : \mathbb{R}^{14} \rightarrow \mathbb{R}^6$$

produces

$$z = E(X).$$

The latent representation z corresponds to the six-factor structural decomposition.

Thus convenience yield admits an AI-style latent-factor interpretation.

14 Algorithm

Pseudo-Code

Input:

14-factor convenience-yield vector

Compute:

Structural projection matrix A

Output:
6-factor structural vector

```

Procedure:
y6 = A*y14

'''
if rank(y6) > 10:
    reject model

return y6
'''

```

15 Statistical Framework

Assume

$$y_t = \beta_0 + \sum_{i=1}^6 \beta_i x_{it} + \varepsilon_t.$$

The structural decomposition can be estimated using

- Principal Components,
- Factor Models,
- State-Space Models,
- Bayesian Inference,
- Machine Learning.

16 Network Representation

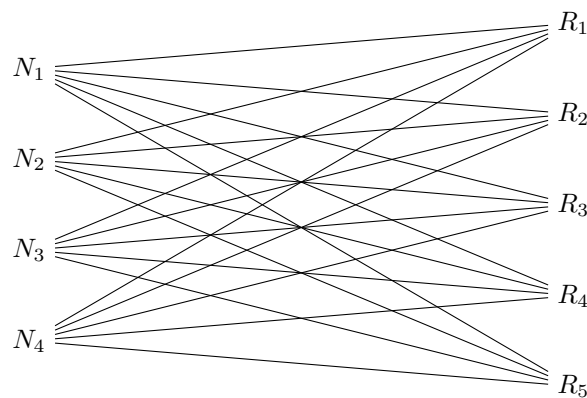


Figure 1: Bounded nation-currency architecture

17 Conclusion

The distinction between phenomenological and structural convenience-yield decompositions is fundamental.

A fourteen-factor decomposition may be useful descriptively.

However, under a bounded nation-currency architecture satisfying

$$N_{\max} \leq 4, \quad R_{\max} = 5,$$

the dimensionality of convenience-yield generation cannot exceed ten.

Consequently, a six-factor structural basis provides a more parsimonious and dimensionally consistent representation.

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Glossary

Convenience Yield Non-monetary benefit from physical possession.

Phenomenological Decomposition A descriptive list of observable components.

Structural Decomposition A basis representation respecting dimensional constraints.

Nation–Currency Pair Interaction between a nation and a reserve-currency system.

Operational Yield Benefits arising from production continuity.

Option Yield Benefits arising from future flexibility.

Risk Yield Benefits arising from uncertainty reduction.

Strategic Yield Benefits arising from security and power.

Informational Yield Benefits arising from superior information.

Political Yield Benefits arising from geopolitical influence.

The End